



Problem Challenge 2

We'll cover the following



- Minimum Height Trees (hard)
- Try it yourself

Minimum Height Trees (hard)

We are given an undirected graph that has characteristics of a **k-ary tree**. In such a graph, we can choose any node as the root to make a k-ary tree. The root (or the tree) with the minimum height will be called **Minimum Height Tree (MHT)**. There can be multiple MHTs for a graph. In this problem, we need to find all those roots which give us MHTs. Write a method to find all MHTs of the given graph and return a list of their roots.

Example 1:

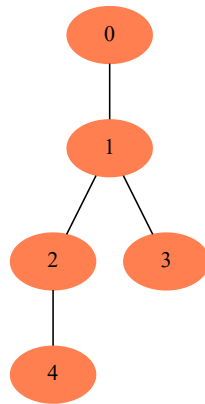
Input: vertices: 5, Edges: [[0, 1], [1, 2], [1, 3], [2, 4]]

Output:[1, 2]

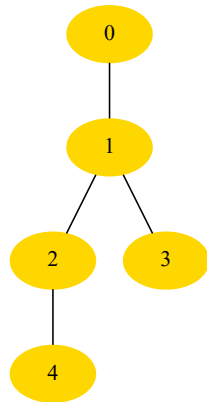
Explanation: Choosing '1' or '2' as roots give us MHTs. In the below diagram, we can see that the height of the trees with roots '1' or '2' is three which is minimum.



Given graph ==>

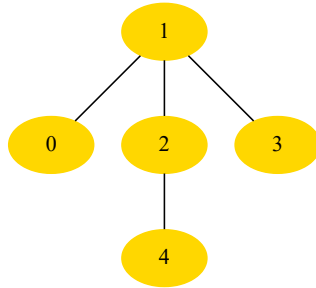


With '0' as the root



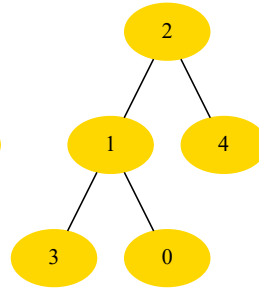
Height = 4

With '1' as the root



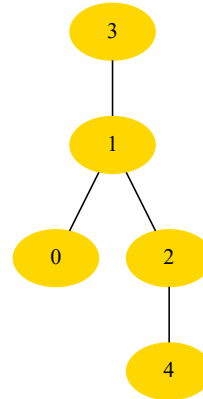
Height = 3

With '2' as the root



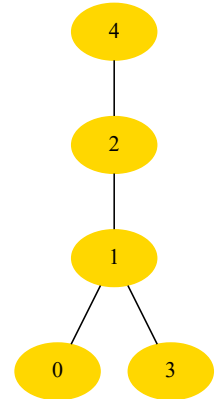
Height = 3

With '3' as the root



Height = 4

With '4' as the root



Height = 4

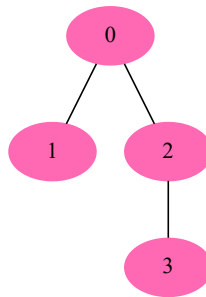
Example 2:

Input: vertices: 4, Edges: [[0, 1], [0, 2], [2, 3]]

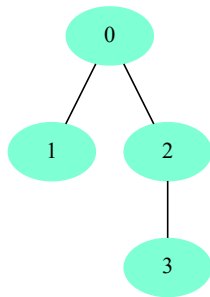
Output:[0, 2]

Explanation: Choosing '0' or '2' as roots give us MHTs. In the below diagram, we can see that the height of the trees with roots '0' or '2' is three which is minimum.

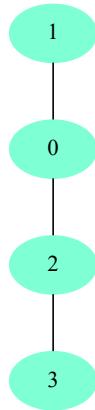
Given graph ==>



With '0' as the root

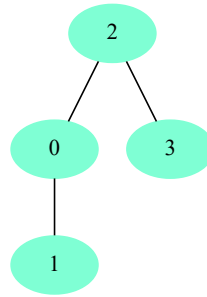
**Height = 3**

With '1' as the root

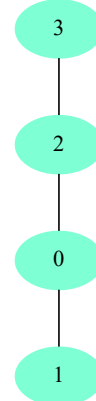


Height = 4

With '2' as the root

**Height = 3**

With '3' as the root



Height = 4